WELLSENSE · TRACEABILITY MATRIX

WellSense traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

WellSense traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-002	PR-001	Measurement range	WLS-CAL-001, WLS-STM-001	Analysis (WLS-CAL-001 [A1], [A3]) and datasheet, TRL 3	Met on paper: 98.07 kPa full scale; swings to 9 m fit with 1 m headroom
UR-002	PR-002	Installation depth	WLS-CAL-001, WLS-BLD-001, WLS-STM-001	Analysis (WLS-CAL-001 [D3], [D4], [D5], [D8]; WLS-STM-001 S11, S13, S14), TRL 3	Met on paper: cable factor 11.5 and tube stress 0.82 MPa at 60 m; the collar carries the tube on the seal plate (0.24 MPa) and a clevis pin above it holds the string if the collar's grip creeps (WLS-DEC-001 D-S1)
UR-001	PR-003	Resolution	WLS-CAL-001, WLS-BLD-001, WLS-STM-001	Analysis (WLS-CAL-001 [A2]), TRL 3	Met on paper: 0.52 mm (1.04 mm on the 20 m variant)
UR-001	PR-004	Accuracy after field calibration	WLS-CAL-001, WLS-BLD-001, WLS-STM-001	Analysis (WLS-CAL-001 [C4], [C4b]), TRL 3; side-by-side with a manual tape at TRL 4	At risk: 16.0 mm root sum square, 37.1 mm worst case, with a 0.25 % class probe (span from a 1 m lift would be about 148 mm at 10 m, so it is not used); a 0.5 % class probe does not meet it (WLS-STM-001 S7 open)
UR-001, UR-014	PR-005	Long-term drift	WLS-CAL-001, WLS-STM-001	Field record, TRL 4	To verify at TRL 4: drift of low-cost transducers is not published; a quarterly check leaves at most 5 mm uncorrected (WLS-STM-001 S10 open)
UR-003	PR-006	Reading interval	WLS-CAL-001, WLS-STM-001	Analysis (WLS-CAL-001 [E3], [E4]) and firmware configuration, TRL 3	Met by design: 30 min batches use 22.7 s a day at SF9, within 30 s fair use
UR-004	PR-007	Local storage	WLS-CAL-001, WLS-STM-001	Analysis (WLS-CAL-001 [E1]), TRL 3	Met on paper: 1.12 MB a year, 6.7 % of FieldNode flash
UR-005	PR-008	Energy	WLS-CAL-001, WLS-BLD-001, WLS-STM-001, WLS-OPT-001	Analysis (WLS-CAL-001 [B2], [B3]), TRL 3	Met on paper: 38.2 mWh a day, 1.6 % of the allowance

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-006	PR-009	Water safety	WLS-CAL-001, WLS-BLD-001, WLS-STM-001, WLS-OPT-001, BOM	Inspection of material certificates, at purchase	Not met on evidence: no certificate in hand for the low-cost cable, seals or probe; the material classes can be certified (WLS-STM-001 S24)
UR-007	PR-010	Compatibility with the existing pump	WLS-CAL-001, WLS-BLD-001, WLS-STM-001, WLS-OPT-001	Design review (WLS-CAL-001 [D1], [D9]), TRL 3; installation trial, TRL 4	At risk: 2.3 mm radial clearance at the nominal bore and 1.24 mm at a cement bead in the thickest pipe, so joints are wiped and a 23 mm drift is pulled through before the probe goes in (WLS-DEC-001 D-S2); fitting a tube may still need the pump pulled
UR-008	PR-011	Installation	WLS-CAL-001, WLS-BLD-001	Installation trial, TRL 4	To verify at TRL 4: 125 min estimate with the locks fitted (WLS-CAL-001 [G1]), 5 min over
UR-009	PR-012	Cost	WLS-CAL-001	Priced BOM (WLS-CAL-001 [H2]), TRL 3	Over target: $264.40 at 30 m for the constructable design with its locks and the clevis pin, $64.40 over; $403.40 with the FieldNode core
UR-010	PR-013	Data use	WLS-CAL-001	Demonstration with sample data, TRL 4	To verify at TRL 4: software not started
UR-011	PR-014	Data ownership and privacy	WLS-CAL-001	Design review, TRL 3	Met by design
UR-012	PR-015	Environment	WLS-CAL-001, WLS-BLD-001, WLS-STM-001	Datasheets (WLS-STM-001 S21), TRL 3; field test later	At risk: met on datasheets for WellSense parts; the junction box runs about 66 °C inside at 55 °C in sun by a first-order balance, but measured enclosures run 20 to 35 K above ambient, near the modules' 85 °C (STM S21 open); the FieldNode core is rated to 45 °C ambient, not 55 °C, so sites above 45 °C are not used until FieldNode confirms (WLS-DEC-001)
UR-013	PR-016	Tamper resistance	WLS-CAL-001, WLS-	Design review, TRL 3	Met by design: cable in galvanized conduit and a short flexible tail; security-head screw on the rim band, padlock on the cap's eye bolt and a

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			BLD-001, BOM		hasp and padlock on the box (BOM line 16); the padlock does not stop the cap being lifted, which the steel cover does at exposed sites
UR-014	PR-017	Maintenance interval	WLS-CAL-001, WLS-STM-001	Analysis (WLS-STM-001 S20; WLS-CAL-001 [F1]), TRL 3	Met on paper: 10 g of gel lasts about 552 days at worst-case daily breathing; permeation through gaskets and glands is not counted
UR-015	PR-018	Extra-low voltage	WLS-CAL-001, WLS-STM-001	Design review (WLS-STM-001 S27, S28), TRL 3	Met by design: 24 V boost output is the highest voltage, within extra-low voltage; a 50 mA resettable fuse protects the shunt (WLS-DEC-001 D-S4)

Gaps

None.